

Dave's Daily

AI • Bayesian optimization US equities

tldr

1. OpenAI is putting GPT-6 in ChatGPT for a claimed 1.2 billion weekly users. Paid plans started Wednesday. Free and Go start Thursday, on a smaller model. On web-search questions, GPT-6 Instant begins answering 44% sooner than GPT-5.6 Instant.
2. Jump Trading says GPT-6 Astra can run a quant study for days. A person still accepts any trading signal. The account is OpenAI's, and it reports no trading result.
3. A new optimizer does the slow statistics while the experiment is still running, then reweights. On lab tasks it keeps the accuracy of fully Bayesian search at about the speed of the usual shortcut.

AI

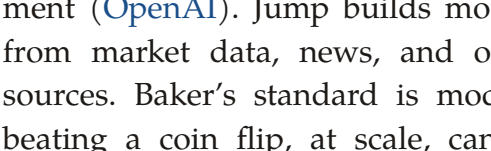
The answer can be a tool. On Wednesday OpenAI began rolling GPT-6 into the Chat tab, with a feature it calls Intelligent UI ([OpenAI](#); [The Verge](#)). The model is trained to mix text with graphics, buttons, forms, and charts, or to stay with plain text when that is the useful answer. A compiler draws the interface as the model writes it, from a library of parts, so the user does not wait for the whole reply. OpenAI's examples include a bicycle diagram with tappable parts, a roast plan whose quantities change with the guest count, and a savings calculator built inside the chat.

Two models, one day apart. Plus, Pro, Business, and Enterprise receive GPT-6 Sol, starting Wednesday. Free and Go receive GPT-6 Luna, starting Thursday. Enterprise stays off until an administrator turns it on. Work and Codex are unchanged. OpenAI says both chat models are tuned for ordinary conversation. The Verge calls Sol the mid-range model and Luna the more efficient one.

Faster to start, not a public proof. For questions that need web search, OpenAI says GPT-6 Instant starts answering 44% sooner, on average, than GPT-5.6 Instant (chart). A separate internal test says GPT-6 Extra High begins answering in the same time as GPT-5.6 Medium and scores above GPT-5.6 Extra High. OpenAI does not publish that score. It also says GPT-6 resisted multi-turn attempts to bypass its safety training more strongly than GPT-5.6 Sol, and points to a [system card](#). A Wednesday briefing report notes that OpenAI did not say whether the pictures raise token use, and that a chart in the reply can still be wrong ([SiliconANGLE](#)). The company says this reaches more than 1.2 billion weekly ChatGPT users. That reach is OpenAI's own figure.

When a web-search answer starts

Index, GPT-5.6 Instant set to 100



Shorter is sooner. The gold bar is this edition's reading of "44% sooner": $100 \times (1 - 0.44) = 56$. OpenAI did not publish the index or the underlying seconds. The axis starts at zero. Source: OpenAI, [GPT-6 for everyone](#), 7 October 2026.

Markets

A trading signal still needs a person. OpenAI's customer story on Jump Trading quotes Lucas Baker, the firm's head of LLM research and development ([OpenAI](#)). Jump builds models from market data, news, and other sources. Baker's standard is modest: beating a coin flip, at scale, can be enough. With GPT-6 Astra he says a study can run for days, pull many sources, decide what matters, and stack small improvements without a person checking every round.

What the story does not show. If an agent emits a trading signal, Baker says it is scoped and reviewed like any other signal: usually informative, possibly wrong, and only then mixed into a controlled execution system. He also says even Astra's longest jobs still check in with the person who set the data, the clock, and what counts as important. The page reports no profit, no fill, and no change in Jump's order flow. News indexes date the page to 6 October. The text retrieved this morning has no date-line. Hypothesis: this is a description of a research desk, not evidence that a model is choosing US equity trades.

Key numbers (this edition)

Item	Figure
Weekly ChatGPT users, OpenAI's figure	1.2 billion
Sooner start, web-search questions	44%
Lab-task latency cut, Matérn kernel	>10×
Deep-kernel latency cut	70–110×
Synthetic tests, input dimension	2 to 10

Bayesian optimization

Scaling: hide the slow step inside the experiment. Fully Bayesian optimization averages a Gaussian process over many hyperparameter settings, which is more honest when data are scarce and more expensive after every new observation. Gustavo Sutter, Alejandro Comas-Leon, and colleagues at the Vector Institute and Waterloo run that sampling while the experiment is still busy ([arXiv:2610.08969](#), 6 Oct). When the result arrives, each sample is reweighted by how well it predicted that result. The next query uses the weighted average. They call it ELF-BO. *The mechanism.* The weight is the predictive density of the new observation under the old posterior, normalized across samples. That identity is Bayes' rule, so the expensive Markov-chain sampling can finish before the measurement does. What remains is a reweight and the usual search for the best next point. If the new result surprises most of the samples, a few weights dominate. They track that with an effective sample size, available as soon as the result arrives.

Data efficiency. On three lab libraries in the Olympus suite (a chromatography sequence, and two polymer blends), fully Bayesian search improved the early part of the run relative to the usual single best-fit of the hyperparameters. ELF-BO tracked that curve. The budget on 15 synthetic functions, from 2 to 10 inputs, was 100 steps after 2*d* starting points, with 64 hyperparameter samples. The paper does not offer a regret proof.

The evidence. On those lab tasks with a Matérn kernel, decision latency — time from a result to the next query — was more than ten times lower than standard fully Bayesian optimization, and close to the single-fit shortcut, with a similar area under the progress curve. With a two-layer neural kernel, standard fully Bayesian latency grew five to nine times versus Matérn, while ELF-BO stayed under a second: a 70 to 110 times cut. On one growing polymer dataset that neural version reached 24 minutes; ELF-BO stayed under three seconds. The miss they flag is Michalewicz in five dimensions, where the effective sample size ran lower and ELF-BO pulled away from the fully Bayesian curve. The latency gain exists only when the experiment outlasts the sampling. Instant simulators do not hide it. They tested Gaussian processes only, including a neural kernel, and name Bayesian neural nets as future work. Hypothesis: the gain is a schedule, not a new model of the objective. Yesterday's paper limited how wiggly a function could be. This one spends idle time on a better average.